Kyle Corbett

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EDUCATION

San Jose State University **San Jose, CA** *May 2025*
Master of Science in Physics | GPA: 3.587
- **Awards/Honors:** Sigma Pi Sigma Honor Society

San Jose State University **San Jose, CA** *May 2023*
Bachelor of Science in Physics | GPA: 3.596
- **Awards/Honors:** Cum Laude, President's Scholar (2x), Dean's Scholar (2x)

SKILLS

Python (NumPy, SciPy, Seaborn, Pandas, PyPlot), SolidWorks, Julia, Rust, Bash, Git, WSL, Linux, Mathematica, LaTeX, HPC, Slurm, Arduino, Raspberry Pi, automotive drivetrain, physics modeling, Monte-Carlo simulation.

PROJECTS

Automotive Design Portfolio

- Designed, prototyped, and finalized a portfolio of over 30 unique tools, replacement parts, and custom solutions for the Z32 and AE86 platforms using SolidWorks, Simplify3D, and 3D printing technology
- Reverse-engineered discontinued, or difficult-to-source OEM parts to restore vehicle functionality.
- Developed methods for testing, iterative prototyping, and improving parts over time to ensure long-term reliability.
- Performed CFD simulation of intake piping and manifolds to improve performance.

Physics Simulations

- Developed Monte Carlo simulations on periodic chain, square, and kagome lattice structures for the Ising, XY, Potts, and Fermi-Hubbard models using Python, Julia, and Rust to study phase transition behavior.
- Created web-based interactive Ising, XY, and Potts simulation applications implementing common Monte-Carlo algorithms using Rust, WebAssembly, HTML, JavaScript, and CSS to teach students about model behaviors.
- Added functionality to the determinant quantum Monte Carlo package SmoQyDQMC for storing results in memory, greatly simplifying file I/O, reducing runtimes by up to 77%, and preventing filesystem-related crashes.
- Simulated quantum computer behaviors using Qiskit to implement common quantum computing algorithms.

PyConsult / PiConsult

- Developed a free, open-source, cross-platform desktop interface for interfacing with 1990s-era Nissan ECUs using the Consult I protocol via RS232 serial, written in Python using NumPy, PySerial, and Tkinter.
- Scaled the desktop interface down to a pocket-sized comprehensive solution for end-user portable diagnostics and data logging using Raspberry Pi, Python, and SPI displays to send, receive, and display data.

WORK EXPERIENCE

Research Assistant **San Jose, CA** *2023-Current*
- Developing Monte Carlo simulations for the Ising, XY, and Fermi-Hubbard models using Julia and Python, with packages Numba, NumPy, SciPy, MatPlotLib, and Jupyter Notebooks on a Slurm-operated HPC.
- Statistical analysis of models near critical temperature with an emphasis on speed, efficiency, and accuracy
- Scaling simulations to tens of thousands of sites and interpreting numbers of data points in the millions.

Teaching Assistant **San Jose, CA** *2023-2025*
- Instruction of Fundamentals of Physics, a non-calculus-based class covering mechanics, heat, and sound.
- Communicated physics concepts to classes of 25 students with little or no physics background.
- Coordinated with up to 25 other instructors and staff to ensure consistency between course sections.

Project Lead **San Jose, CA** *2024*
- Led a team of 6 undergraduate students in developing a comprehensive solution manual of over 100 pages for SJSU's Fundamentals of Physics course to aid new teaching assistants.
- Formulated a structure and procedure for ensuring quality and accuracy in student work

Instructional Student Assistant **San Jose, CA** *2023-2025*
- Supported students in undergraduate Modern Optics, graduate-level Electricity and Magnetism, and Mathematical Methods classes by hosting office hours and providing meaningful feedback on student work.